

## **Data Preparation Session 4**

### **Exercises**

#### **Exercise 4.1**

- a) Import the data-messy.xlsx file as rainfall. Pay attention to the right header.
- b) Separate the month and the period into two columns.
- c) Remove mm from the data where it is necessary and change the data type (to prepare for later analysis).
- d) Restructure data, the geographical location should be a variable (call it "Place"). The new numeric variable should be called "rainfall".
- e) Take a look at the database now. Propose an additional change (and explain shortly why) and execute it.

#### **Exercise 4.2**

- a) Read the coffeeshop.csv file to a data frame as it is using the tidyverse library. Do not change variable types and names.
- b) The data frame is not tidy. Explain what is wrong and why.
- c) Using pivot functions, restructure the data frame.
- d) Clean the first column, it should be named 'Beverage\_category' and should have only text, but no ... or numbers.
- e) Check the unique values of Total fat and Caffeine. What problems do you see with them?

#### **Exercise 4.3**

Go back to Exercise 3.2 and solve the remaining problems with the athletes' data.

#### **Exercise 4.4**

Go back to Exercise 3.6 and solve the remaining problems with the sales data.